



How to Run NexCare on Your Computer

Complete step-by-step guide — no prior experience needed

Beginner Friendly

Spring Boot + Angular

localhost:4200

0 Understanding the Basics First

WHAT IS LOCALHOST?

When you run a website on your own computer, it doesn't go on the internet — it runs privately on your machine. "localhost" is just a name your computer uses to refer to itself. The number after it (like :4200 or :8080) is the port — think of it as a door number in a building. Port 4200 is where you view the app. Port 8080 is where the backend (data) lives behind the scenes.

WHAT IS A TERMINAL / COMMAND PROMPT?

A Terminal is a text-based window where you type commands to control your computer. On Windows it's called "Command Prompt" or "PowerShell". On Mac it's called "Terminal". You don't need to be an expert — you will just copy and paste the commands from this guide.

WHY DO WE NEED TWO TERMINALS?

NexCare has two parts running at the same time: the **backend** (Java/Spring Boot, handles data) and the **frontend** (Angular, the visual website). Each needs its own terminal window to run. Think of it like two engines in a car — both must be on for the car to drive.

1 Install the Required Software

You need to install these **four tools** before running the project. Do this only once — you won't need to repeat it.

A Install Java 17 (JDK)

Java is needed to run the backend (Spring Boot).

1. Go to: <https://adoptium.net>
2. Click "**Latest LTS Release**" → select **Temurin 17**
3. Download the installer for your OS (Windows .msi or Mac .pkg)
4. Run the installer and click Next → Next → Finish
5. To verify it worked, open a terminal and type:

```
java --version
```

You should see something like: `openjdk 17.0.x ...`

B Install Maven

Maven is a build tool for Java projects.

1. Go to: <https://maven.apache.org/download.cgi>
2. Download "**Binary zip archive**" (the .zip file)
3. Extract it to a folder like `C:\Program Files\Maven`
4. Add Maven's `bin` folder to your system PATH environment variable
5. Verify:

```
mvn --version
```



Easier option (Mac): If you have Homebrew, just run `brew install maven` in a terminal.

C Install Node.js

Node.js is needed to run the Angular frontend.

1. Go to: <https://nodejs.org>
2. Click the **LTS** version (the left green button)
3. Run the installer — click Next all the way through
4. Verify:

```
node --version    # should show v18 or higher
npm --version     # should show 9 or higher
```

D Install Git

Git is needed to download the project from GitHub.

1. Go to: <https://git-scm.com/downloads>

2. Download and run the installer for your OS

3. Verify:

```
git --version
```

2 Download the Project

How to open a terminal

On Mac: Press `Cmd + Space`, type "Terminal", press Enter

On Windows: Press `Win key`, type "Command Prompt" or "PowerShell", press Enter

Download (clone) the project

In the terminal, type these commands one by one and press Enter after each:

```
# Go to your Desktop (or wherever you want to save the project)
cd Desktop

# Download the project from GitHub
git clone https://github.com/rishithamanyam/Capstone-Project.git

# Go into the project folder
cd Capstone-Project/SIMS
```



You now have the full project on your computer inside a folder called **Capstone-Project**.

Open in VS Code

Type this command in the same terminal:

```
code .
```

This opens the entire project in VS Code. (If `code .` doesn't work, open VS Code manually → File → Open Folder → select the SIMS folder.)

3 Start the Backend (Spring Boot)



Make sure you are in the **SIMS/** folder in your terminal, not in any subfolder.

Type this command and press Enter:

```
# This downloads all Java libraries and starts the server  
mvn spring-boot:run
```



First time only: Maven will download many libraries. This can take 2–5 minutes. Your internet must be on. You'll see lots of text scrolling — that's normal.

When the backend is ready, you will see this line in the terminal:

```
2024-01-15 10:23:45.123 INFO --- [main] c.w.c.CsdApplication:  
Started CsdApplication in 3.421 seconds (JVM running for 4.012)
```



The backend is now running on port 8080. **Leave this terminal open.** Do not close it while using the app.



VS Code shortcut: Instead of typing the command, press **Ctrl+Shift+B** (or **Cmd+Shift+B** on Mac) and select **Run NexCare (Full Stack)** — this starts both backend AND frontend automatically.

4 Start the Frontend (Angular)



Open a **new, second terminal window** for this step. Do NOT close the first terminal.

In the new terminal, navigate to the frontend folder:

```
# Go into the frontend folder  
cd Desktop/Capstone-Project/SIMS/frontend  
  
# Install frontend packages (first time only – takes ~1 minute)  
npm install
```

```
# Start the Angular development server  
npx ng serve --port 4200 --open
```

When Angular is ready, you will see:



```
Building...
```

```
✓ Compiled successfully.
```

```
** Angular Live Development Server is listening on localhost:4200 **
```



The browser will open automatically at **<http://localhost:4200>**. You should see the NexCare login page!

5 Using the App

The browser will show the NexCare landing page. Click **Sign In** and use these credentials:

Role	Email	Password	What you can do
ADMIN	admin@csd.com	admin123	Manage employees, all tickets, outages, analytics
MANAGER	john.smith@csd.com	Welcome@123	View team tickets and outages
REP	mike.rep@csd.com	Welcome@123	Update ticket statuses, search customers
CUSTOMER	alice@gmail.com	Customer@123	Raise tickets, check plans, live chat



To logout and switch to a different account, click the user avatar (initials) at the bottom of the left sidebar.

Navigation

Use the sidebar on the left. Click any item — the page updates instantly without reloading (because Angular is a Single Page Application).

For the **Admin** dashboard specifically, clicking sidebar items switches between tabs: Dashboard, Employees, Tickets, and Outages.

6 How to Stop the App

When you're done, stop both terminals:

1 Stop the Angular server (Terminal 2)

Click inside Terminal 2 and press **Ctrl + C** on your keyboard. Type **Y** if asked to confirm.

2 Stop the Spring Boot server (Terminal 1)

Click inside Terminal 1 and press **Ctrl + C** on your keyboard.



VS Code shortcut: Run Task → "🛑 Stop All Servers" to stop both at once.

7 If Something Goes Wrong



"mvn: command not found" or "mvn is not recognized"

Maven is not installed properly. Re-do Step 1B and make sure you added Maven to your system PATH.



Port 8080 already in use

Another app is using port 8080. On Mac, run `sudo lsof -i :8080` then `kill -9 [PID]`. On Windows, run `netstat -ano | findstr :8080` then `taskkill /PID [number] /F`.



Login says "Invalid credentials" even with demo accounts

The database may already have data from a previous run. Delete the `csddb.mv.db` file in the SIMS folder and restart the backend. It will re-create fresh demo data.



Browser shows blank white page

Angular may still be compiling. Wait 30 seconds then refresh. If still blank, check Terminal 2 for error messages.



"npm install" fails

Delete the `node_modules` folder inside `frontend/` and run `npm install` again. Make sure your internet connection is working.

8 Quick Pre-Demo Checklist

Run through this list 10 minutes before your presentation:

- ☐ Java 17 is installed (java --version shows 17+)
- ☐ Maven is installed (mvn --version works)
- ☐ Node.js is installed (node --version shows 18+)
- ☐ Terminal 1 is open and backend is running (shows "Started CsdApplication")
- ☐ Terminal 2 is open and Angular is running (shows "Compiled successfully")
- ☐ Browser is open at <http://localhost:4200>
- ☐ Logged in as Admin to show dashboard is working
- ☐ Demo credentials are written down or memorised
- ☐ Internet is not required once both terminals are running